



Quasi-solid Dye Sensitized Solar Cells Having Straight Ion Paths

Proposal of High-Efficiency QDSCs Consisting of Two Carrier Density Layers

Takeshi Kogo,^a Shuzi Hayase,^{a,*} Tatsuo Kaiho,^b and Mitsuru Taguchi^b

^aKyushu Institute of Technology, Wakamatsu-ku, Kitakyushu, Fukuoka 808-1096, Japan

^bKanto Natural Gas Development Company, Limited, Mobara, Chiba 297-8550, Japan

Quasi-solid dye-sensitized solar cells (QDSCs) consisting of amylose/porous Al_2O_3 film/ionic liquid films (Am-Al/IL film) are reported. The surface of straight nanopores in the Al_2O_3 film was modified with amylose and these nanopores were filled with ionic-liquid-type electrolytes. The ionic liquid did not leak from these nanopores of the film. The photovoltaic performance of the QDSC consisting of Am-Al/IL films was almost the same as that before the solidification, despite that the film was occupied with 50% of electrochemically inactive Al_2O_3 wall. It was found that the iodine was adsorbed on the straight wall of Am-Al films and the apparent diffusion coefficient of I_3^- species was almost the same as that in the parent liquid electrolyte. From these results and our previous reports, we propose QDSC structures consisting of two-carrier-density layers (I^-/I_3^-) with low and high carrier concentrations for high-efficiency cells. A bulk area between a counter electrode and a TiO_2 electrode is occupied with a high-carrier-density medium (I^-/I_3^-), and TiO_2 nanopores are filled with a low-carrier-density medium (I^-/I_3^-). The former increases the conductivity and the latter suppresses the back electron transfer from the TiO_2 layer to redox species.

© 2008 The Electrochemical Society. [DOI: 10.1149/1.2955722] All rights reserved.

Manuscript submitted March 31, 2008; revised manuscript received May 27, 2008. Published July 25, 2008.

Dye-sensitized solar cells (DSCs) have attracted interest as low-cost solar cells with high efficiency.¹ Since Grätzel et al. reported the DSC structure, efforts to increase the efficiency have been made and about 11% efficiencies have been reported so far.^{2,3} Another research interest on the DSC is solidification. All-solid-state DSCs having p-type organic^{4,7} and inorganic^{8,9} semiconductive mediums have been reported. Another approach is to use gel electrolytes. Considering the fact that suppressing volatility and avoiding leakage of liquid electrolytes are the purpose of the solidification, quasi-solidification of electrolytes consisting of nonvolatile ionic-liquid-based electrolytes meets the requirement. Ionic liquids have been known to be nonvolatile electrolytes, and the application to DSCs has been tried.¹⁰⁻¹⁷ Ionic-liquid-based gel electrolytes have been intensively studied.¹⁸⁻²⁴ In order to obtain the gel material, gelators (solidification materials) have to be added to the ionic-liquid-based electrolyte. Polymer-based gelators and nanomaterial-based gelators have been reported.¹⁸⁻²⁴ Because gelators are electrochemically inactive, ionic diffusion of I_3^- is suppressed and short circuit current (J_{sc}) decreases with an increase in the amount of gelator added to the electrolytes. Therefore, the amount of gelators has been limited to around 10 wt % vs electrolytes in order to maintain the photovoltaic performance. If the amount of gelators is increased in order to make harder charge-carrier mediums (high-solid mediums), I_2 concentration has to be increased, because swift ionic diffusion is suppressed in the high-solid medium. However, the increase in I_2 concentration increases the chance of charge recombination between electrons in TiO_2 layers and I_2 in electrolytes, resulting in decreasing open-circuit voltage (V_{oc}).

We have already reported a composite I^-/I_3^- conductive film with straight pores.²⁵ The composite film consists of surface-modified Al_2O_3 (S-Al) having straight pores and ionic liquids (S-Al/IL film), where the straight nanosurface was chemically modified with self-assembled molecules substituted with methylimidazolium iodide moieties.²⁵ In the nanopore surfaces of S-Al films, I^-/I_3^- species were concentrated and made ionic paths. The photovoltaic performance of the cell consisting of the S-Al/IL film was unexpectedly better than that of the cell consisting of the parent-liquid-type electrolyte, despite the fact that the content of the electrochemically inactive alumina wall was 50% (high-solid medium). We explained

the higher apparent ionic diffusion of I_3^- species in the S-Al/IL film as a Grotthius mechanism.²⁵ This prompted us to propose a general structure for quasi-solid DSC (QDSC) with high efficiencies. In this paper, we propose a QDSC consisting of two carrier density layers (I^-/I_3^-), as shown in Fig. 1. A bulk area (A-2) between a counter electrode and a TiO_2 electrode is occupied with straight charge carrier mediums (I^-/I_3^-) of high carrier density, and TiO_2 nanopores (A-1) are filled with mediums (I^-/I_3^-) of low carrier density. The former increases the conductivity and the latter suppress the back electron transfer from the TiO_2 layer to redox species. In this paper we report a QDSC consisting of amylose as the material giving the high-density I^-/I_3^- medium. In addition, we want to prove the effectiveness of the cell 1 structure by using QDSC consisting of amylose as well as that consisting of an S-Al/IL film previously reported.

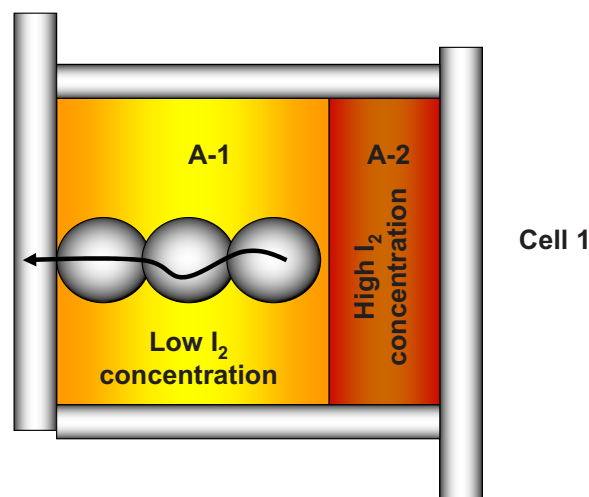


Figure 1. (Color online) Proposed QDSC, cell 1, consisting of two-carrier-density layers of low and high carrier concentrations. A-1: Nanopores in TiO_2 layers is filled with low- I_2 -density mediums which suppress back electron transfer. A-2: Bulk layer between a TiO_2 electrode and a counter electrode is occupied with high- I_2 -density mediums which bring about high conductivity.

* Electrochemical Society Active Member.

^z E-mail: hayase@life.kyutech.ac.jp

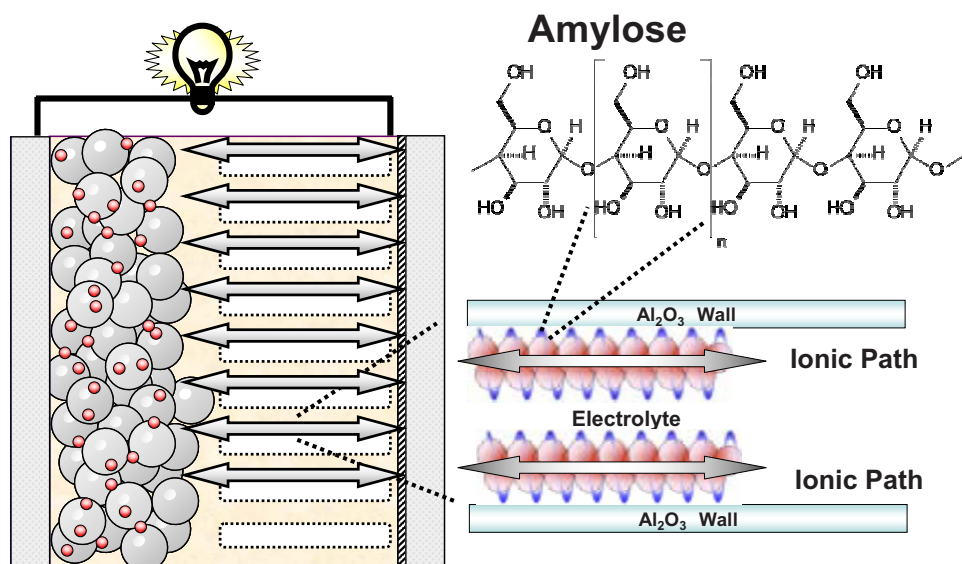


Figure 2. (Color online) QDSC structure consisting of porous alumina treated with amylose. Amylose makes complexes with iodine as shown in the figure.

Experimental

Material.— A F/SnO₂ glass (Low E glass) was purchased from Nippon Sheet Glass Co., Ltd. TiO₂ paste (Ti-nanoxide D) and dye [*cis*-bis(isothiocyanato) bis(2,2'-bipyridyl-4,4'-dicarboxylato)-ruthenium(II) bis-tetrabutylammonium (ruthenium 535 bis-TBA, N719)] were purchased from Solaronix S.A. I₂, LiI, acetonitrile, and *t*-butyl alcohol were commercially available from Wako Chemicals Co., Ltd.

Amylose solution (Glico Co., Mw 200 k) was solved in hot water, and 0.05, 0.1, 0.5, 1.0, and 2.0 wt % aq. solutions were prepared. Al₂O₃ films (thickness 65 μm, pore diameter 200 nm, Whatmann) were baked at 450°C before use. The porous alumina film was dipped in the amylose solution and dried as it was. The film is abbreviated as Am-Al film.

Fabrication of solar cells.— Ti-nanoxide D paste was coated on a low E glass with a doctor blade. The substrate was baked at 450°C to fabricate TiO₂ layers of 12 μm thickness. The substrate was dipped in the solution containing ruthenium 535 bis-TBA. The concentration was fixed to 0.3 mM in *t*-butyl alcohol and acetonitrile (1:1 vol/vol). A Pt-sputtered SnO₂/F layered glass substrate was employed as the counter electrode. The Am-Al film was put on the TiO₂ substrate, and ionic liquid type electrolyte (IL-1) was dropped on the Am-Al film. After that, the counter electrode was set on the Am-Al film. A Kapton tape (Permacel, Nitto Denko Co.) with 65 μm thickness was used as the spacer. The cell area was 0.25 cm². The solar cell measurement was carried out by using a photomask on the solar cell in order to remove the effect of optical reflection. Electrolyte composition (IL-1) was as follows: LiI: 500 mM, I₂: 300 mM, *t*-butylpyridine: 580 mM, in the mixture of 1-methyl-3-propyl-imidazolium iodide and 1-ethyl-3-methyl-imidazolium tricyanomethanide (3:7).

Photovoltaic performance was monitored with Bunko-Keiki Co., Ltd., model solar simulator KHP-1 equipped with xenon lamp (XLS-150A). The exposure light was adjusted to be air mass 1.5 (100 mW/cm²). UV-vis (UV-visible) spectra of iodine were monitored by a Jasco model V-530 UV-vis spectrophotometer.

The apparent diffusion coefficient was measured by limiting current methods. The electrode size was 0.25 cm². A symmetrical cell was fabricated by sandwiching a porous Al₂O₃ film with two Pt electrodes fabricated on SnO₂/F layered glass substrates. When one Pt electrode is charged negatively, I₃⁻ diffuses to near the electrode and is reduced to I⁻. When the potential is increased negatively, the current increases linearly and levels off at a certain voltage. This current is called limiting current. I₃⁻ diffusion is slower than that of

I⁻ because I₃⁻ is larger than I⁻. Therefore, I₃⁻ is responsible for the limiting current. The apparent diffusion coefficient was calculated by using the formula^{12,15}

$$D = I_{\text{lim}} d / 2nFC$$

where D is the diffusion coefficient, I_{lim} is the limiting current, d is the cell gap, n is the number of electrons (two), F is the Faraday constant, and C is the initial I₂ concentration.

Results and Discussion

We focused on amylose in order to concentrate I⁻/I₃⁻ redox species in straight pores of Al₂O₃ films (Fig. 2) because it is well known that amylose makes complex with I₂.²⁶⁻²⁸

Uchida et al. have reported electrolytes consisting of cyclodextrine molecules and have reported that charge recombination was suppressed, probably because of the interaction between cyclodextrine and iodine.²⁹ However, they have not reported the solidification.

In order to increase I⁻/I₃⁻ density in nanopores of the alumina film with straight pores, the alumina nanopore walls were coated with amylose (Am-Al). The film was dipped in acetonitrile solution containing iodine (3 mM) and LiI (5 mM) for 1 day. Figure 3 shows UV-vis absorption curves of supernatant solutions before and after the Am-Al substrate was dipped in the solution. After the substrate

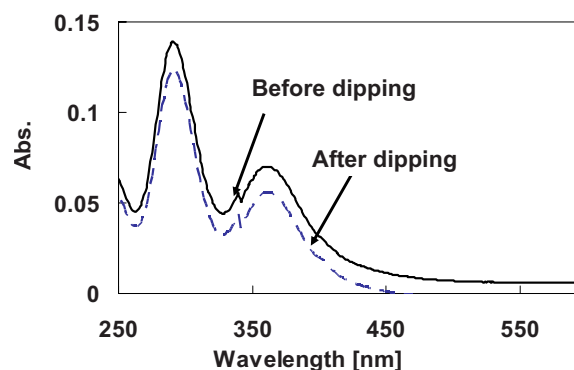


Figure 3. (Color online) UV-vis absorption spectra of electrolyte solutions before and after Am-Al film was dipped. The Am-Al film was dipped in acetonitrile solution containing iodine (3 mM) and LiI (5 mM) for 1 day. The UV-vis spectra of supernatant solutions were measured.

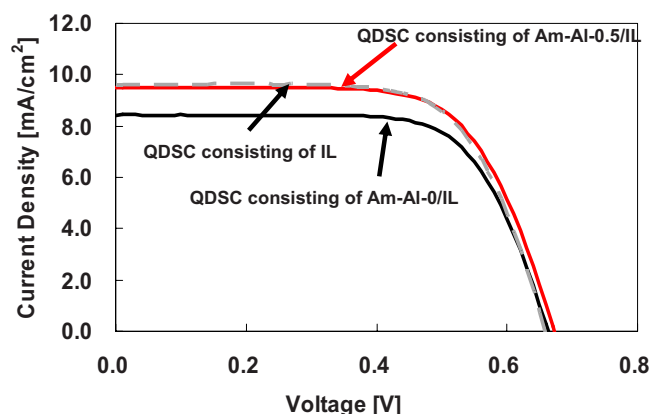


Figure 4. (Color online) Photovoltaic properties of QDSCs. Am-Al-0.5/IL: a porous alumina film surface-treated with 0.5% amylose solution in acetonitrile. The pore was filled with IL-1 electrolyte (see Experimental section); AM-Al-0/IL: a porous alumina film without surface treatment. The pore was filled with IL-1, IL: liquid electrolyte (IL-1) without porous alumina film.

was dipped, the peak intensity of UV-vis absorption of the supernatant solution decreased, suggesting that the iodine was adsorbed and concentrated on the Am-Al film.

Figure 4 shows the relationship between photocurrent and voltage before and after the Am-Al film was inserted between a TiO_2 electrode and a counter electrode. J_{sc} and V_{oc} values of QDSC consisting of Am-Al films treated with 0.5% amylose solution (Am-Al-0.5 film) were almost the same as that of the cell consisting of IL-1 liquid electrolyte, strongly demonstrating that the quasi-solidification was carried out without decreasing photovoltaic performance, despite the higher gelator (alumina) content (50%). In addition, the short-circuit current (J_{sc}) of the QDSC consisting of an Am-Al-0.5 film was higher than that of QDSCs consisting of a nontreated Am-Al-0 film. This strongly demonstrated that the presence of the amylose is crucial for the higher photovoltaic performance.

Table I summarizes photovoltaic properties of QDSCs consisting of Am-Al/IL films, where the Al_2O_3 films were treated with different concentrations of amylose solution. Open-circuit voltage (V_{oc}) and fill factor (ff) did not change largely when the porous alumina was treated with amylose solutions with different concentrations. QDSC consisting of Am-Al-2.0 did not show right current-voltage curves, probably because of insufficient charge carrier diffusion. The efficiency change was associated with the change J_{sc} . Figure 5 shows the relationship between amylose concentration used for the

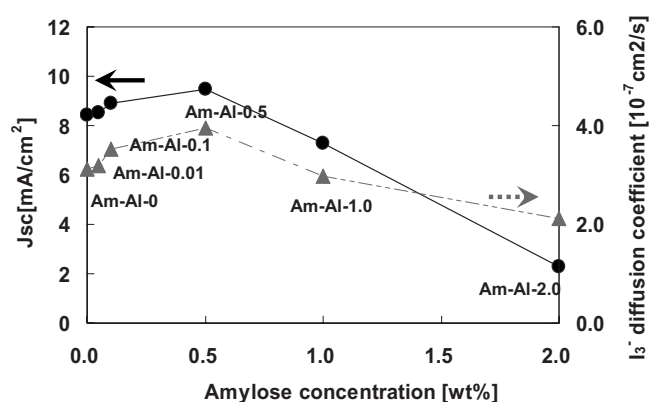


Figure 5. Relationship between J_{sc} and concentration of amylose solution in which porous alumina film was treated. Am-Al-X stands for a film surface-treated with X wt % amylose solution. For example, Am-Al-0.1 stands for a film surface treated with 0.1 wt % amylose solution in acetonitrile. How to measure ionic diffusion was described in the Experimental section.

surface modification and J_{sc} , as well as the relationship between amylose concentration and apparent I_3^- diffusion coefficient in Am-Al/IL film. Ionic diffusion coefficients in Am-Al/IL film treated with 0.5% amylose solution were $3.95 \times 10^{-7} \text{ cm}^2/\text{s}$, which was almost the same as that of the parent-liquid-type electrolyte (IL-1). We have already reported that I_5^- structures were observed in S-Al/IL films. I_5^- has larger consecutive structures of iodide and iodine than I_3^- observed commonly in liquid-type electrolytes, which strongly supported the idea that iodide and iodine are concentrated in the film and charges are carried by the Grotthuss mechanism.²⁵ The same behaviors have been reported on TiO_2 nanoparticles.^{19,30} It is expected that the same phenomena occur in Am-Al/IL films. The ionic diffusion coefficient and J_{sc} increased as amylose concentration for treating alumina films increased to 0.5%, as shown in Fig. 5. After that they were gradually decreased as the concentration increased. The I_3^- diffusion coefficient behaves in the same way as J_{sc} vs the variation of amylose concentration, demonstrating that J_{sc} is closely associated with the I_3^- diffusion coefficient in Am-Al/IL films. The increase in the I_3^- diffusion coefficient in Am-Al films from Am-Al-0 to Am-Al-0.5 (Fig. 5) is explained by the formation of the straight ion paths in Am-Al/IL films. The decrease in the I_3^- diffusion coefficient from Am-Al-1.0 film to Am-Al-2.0 film (Fig. 5) is explained by the total blocking of the nanopores by amylose. The ionic path in Am-Al films is considered to be interrupted in many portions, and I_3^- diffusion at the interrupted portion is helped with I_3^-/I^- redox species in ionic liquids filling the straight nanopores. Namely, Am-Al/IL films have two conduction mechanisms. One is carrier diffusions using straight ion paths in amylose, and the other is common I^-/I_3^- diffusion in ionic liquids filling the nanopores. In order to obtain the best results, appropriate surface modification with amylose was needed.

Conclusion

It was proved that Am-Al/IL films were useful for quasi-solidification of DSCs with high efficiency, where I^-/I_3^- species were concentrated on the nanopores and made ionic paths. The photovoltaic performance of the QDSC consisting of Am-Al/IL films was almost the same as that of DSCs with the parent liquid electrolytes, despite that the film contained 50% of electrochemically inactive alumina wall. Am-Al/IL films have two conduction mechanisms. One is carrier diffusion using straight ion paths in amylose, and the other is common I^-/I_3^- diffusion in ionic liquids filling the nanopores. I_3^- species were concentrated in the nanopores of Am-Al/IL films, and the concentration of I_3^- in nanopores in TiO_2 layers was

Table I. Relationship between J_{sc} and amylose concentration in which porous alumina film was treated.^a

Abbreviation	Am concentration wt %	η (%)	J_{sc} (mA/cm ²)	V_{oc} (V)	FF
IL	Liquid	4.29	9.59	0.66	0.68
Am-Al-0	0	3.87	8.43	0.66	0.69
Am-Al-0.05	0.05	3.71	8.17	0.66	0.69
Am-Al-0.1	0.1	4.27	9.47	0.66	0.68
Am-Al-0.5	0.5	4.32	9.51	0.65	0.69
Am-Al-1.0	1.0	3.65	7.54	0.65	0.75
Am-Al-2.0	2.0	—	1.87	0.63	—

^a Am-Al-X stands for a film surface-treated with X wt % amylose solution. For example, Am-Al-0.1 stands for a film surface treated with 0.1 wt % amylose solution in water. IL stands for ionic liquid type electrolyte IL-1 (see Experimental section). 0.25 cm², AM 1.5, 100 mW/cm².

relatively decreased. The QDSC consisting of Am-Al/IL films proves the effectiveness of the cell 1 structure in Fig. 1.

Kyushu Institute of Technology assisted in meeting the publication costs of this article.

References

1. B. O'Regan and M. Grätzel, *Nature (London)*, **353**, 737 (1991).
2. S. Ito, P. Liska, P. Comte, R. Charvet, P. Péchy, U. Baéché, L. Schmidt-Mende, S. M. Zakeeruddin, A. Kay, M. K. Nazeeruddin, et al. *Chem. Commun. (Cambridge)*, **2005**, 4351.
3. Y. Chiba, A. Islam, Y. Watanabe, R. Komiya, N. Koide, and L. Han, *Jpn. J. Appl. Phys., Part 2*, **45**, L638 (2006).
4. L. Schmidt-Mende, S. M. Zakeeruddin, and M. Grätzel, *Appl. Phys. Lett.*, **86**, 013504 (2005).
5. N. Fukuri, N. Masaki, T. Kitamura, Y. Wada, and S. Yanagida, *J. Phys. Chem. B*, **110**, 25251 (2006).
6. J. Xia, N. Masaki, M. Lira-Cantu, Y. Kim, K. Jiang, and S. Yanagida, *J. Am. Chem. Soc.*, **130**, 1258 (2008).
7. N. Ikeda and T. Miyasaka, *Chem. Commun. (Cambridge)*, **2005**, 1886.
8. Q.-B. Meng, K. Takahashi, X.-T. Zhang, I. Sutanto, T. N. Rao, O. Sato, and A. Fujishima, *Langmuir*, **19**, 3572 (2003).
9. G. R. A. Kumara, A. Konno, K. Shiratsuchi, J. Tsukahara, and K. Tennakone, *Chem. Mater.*, **14**, 954 (2002).
10. N. Yamanaka, R. Kawano, W. Kubo, T. Kitamura, Y. Wada, M. Watanabe, and S. Yanagida, *Chem. Commun. (Cambridge)*, **2005**, 740.
11. F. Mazille, Z. Fei, D. Kuang, D. Zhao, S. M. Zakeeruddin, M. Grätzel, and P. J. Dyson, *Inorg. Chem.*, **45**, 1585 (2006).
12. D. Kuang, P. Wang, S. Ito, S. M. Zakeeruddin, and M. Grätzel, *J. Am. Chem. Soc.*, **128**, 7732 (2006).
13. R. Kawano and M. Watanabe, *Chem. Commun. (Cambridge)*, **2005**, 2107.
14. X. Xiao, X. Zhou, X. Li, Y. Lin, and M. Wang, *Sol. Energy Mater. Sol. Cells*, **91**, 785 (2007).
15. P. Wang, S. M. Zakeeruddin, R. H. Baker, and M. Grätzel, *Chem. Mater.*, **16**, 2694 (2004).
16. P. Wang, S. M. Zakeeruddin, J. E. Moser, and M. Grätzel, *J. Phys. Chem. B*, **107**, 13280 (2003).
17. R. Kawano, H. Matsui, C. Matsuyama, A. Sato, M. A. B. H. Susan, N. Tanabe, and M. Watanabe, *J. Photochem. Photobiol., A*, **164**, 87 (2004).
18. N. Mohmeyer, D. Kuang, P. Wang, H. W. Schmidt, S. M. Zakeeruddin, and M. Grätzel, *J. Mater. Chem.*, **16**, 2978 (2006).
19. W. Kubo, S. Kambe, S. Nakade, T. Kitamura, K. Hanabusa, Y. Wada, and S. Yanagida, *J. Phys. Chem. B*, **107**, 4374 (2003).
20. J. H. Kim, M. S. Kang, Y. J. Kim, J. Won, N. G. Park, and Y. Kang, *Chem. Commun. (Cambridge)*, **2004**, 1662.
21. P. Wang, S. M. Zakeeruddin, I. Exnarb, and M. Grätzel, *Chem. Commun. (Cambridge)*, **2002**, 2972.
22. E. Stathatos, P. Lianos, S. M. Zakeeruddin, P. Liska, and M. Grätzel, *Chem. Mater.*, **15**, 1825 (2003).
23. R. Kumar, A. K. Sharma, V. S. Parmar, A. C. Watterson, K. G. Chittibabu, J. Kumar, and L. A. Samuelson, *Chem. Mater.*, **16**, 4841 (2004).
24. P. Wang, S. M. Zakeeruddin, P. Comte, I. Exnar, and M. Grätzel, *J. Am. Chem. Soc.*, **125**, 1166 (2003).
25. T. Kato and S. Hayase, *J. Electrochem. Soc.*, **154**, B117 (2007).
26. I. A. Jideani, Y. Takeda, and S. Hizukuri, *Cereal Chem.*, **73**, 677 (1996).
27. M. Hirai, T. Hirai, and T. Ueki, *Polymer*, **35**, 2222 (1994).
28. X. Yu, C. Houtman, and R. H. Atalla, *Carbohydr. Res.*, **292**, 129 (1996).
29. S. Uchida, M. Tomiha, Y. Sanehira, and Y. Fushimi, *Kobunshi Ronbunshu*, **63**, 62 (2006).
30. W. Kubo, K. Murakoshi, T. Kitamura, S. Yoshida, M. Haruki, K. Hanabusa, H. Shirai, Y. Wada, and S. Yanagida, *J. Phys. Chem. B*, **105**, 12809 (2001).